Siddhartha Shankar

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Summary

3rd-year CS student at MIT Bengaluru (GPA 9.14/10) with a research-first approach to building. **First-authored** a paper at **ICONICA** on emergent modularity in transformer attention; currently developing a federated GNN for drug-drug interaction prediction. **Gold-level WorldQuant Brain consultant** (top ∼1% globally) — bringing the same depth to quantitative alpha design as to ML systems research.

Skills

Languages: Python, C, C++, Java, JavaScript, SQL

ML & Research: PyTorch, GraphSAGE, Flower, W&B, Hydra

Systems & Backend: Linux, POSIX API, Node.js, Express, Flask, Spring Boot, Playwright, Git

Databases: PostgreSQL, MySQL, MongoDB, Redis

Education

Manipal Institute of Technology, Bengaluru — Expected Jun 2028
Bachelor of Technology in Computer Science — *GPA: 9.14 / 10*

Experience

Research Consultant — 2026 – Present
WorldQuant Brain
- Earned **Gold-level Consultant** status — invite-only designation requiring 10,000+ platform points, placing in the top ∼1% of global contributors.
- Designed and backtested **15+ quantitative alpha signals** for global equity markets; improved out-of-sample Sharpe ratios across multiple datasets using cross-sectional and time-series factor methods.

Quality Assurance Engineer Intern — Jun – Aug 2025
FalkonData
- Designed **100+ test cases** spanning edge conditions, boundary inputs, and integration paths; identified root-cause patterns across recurring failure classes to improve component testability and reduce defect escapes in downstream systems.
- Authored end-to-end automation test suites using **Playwright**; covered critical user journeys across multiple environments to catch regressions before release.

Research

Emergent Modularity in Transformer Attention — *First-authored · ICONICA*
- Large-scale empirical study across **10,706 attention head pairs** (GPT-2 and EleutherAI Pythia families); showed head-level modularity emerges with scale — non-additivity decreases significantly at larger parameter counts — validated via t-tests and ablation studies.

FedDDI-GNN — Federated Drug-Drug Interaction Prediction — *2026 – Present*
- Built a **GraphSAGE + Hadamard MLP** link predictor on **ogbl-ddi** (0.80+ Val Hits@20 centralized); implemented **FedAvg** and **FedProx** baselines under the Flower federated simulation framework.
- Modeled non-IID hospital heterogeneity via ATC-aware Dirichlet sampling ($\alpha = 0.5$); designed concept drift simulator (sudden + gradual modes) tracking nadir degradation and rounds-to-recovery metrics across 100-round FL runs.
- Shipped **101-test** suite with GitHub Actions CI, Hydra config management, and W&B integration; quantified ∼0.50 Hits@20 federation overhead as the baseline the research targets.

Projects

Custom Memory Allocator | — *2025*
- Implemented `malloc/free` from scratch using `sbrk` (small allocs) and `mmap` (>128 KB), replicating glibc behavior; designed block splitting, coalescing, and free-list management; **10,000 sequential alloc/free cycles in under 0.1 ms**.

Shortify | — *2025*
- Built a URL shortener using **Node.js, Express, and MySQL** with deterministic URL deduplication and Base62 short-code generation; benchmarked at **3,000+ HTTP redirects/sec** under **100 concurrent users** with **43.9 ms p95 latency** and **0% request failures**.

Linene | — *2025*
- Full-stack note-taking app (Node.js, MongoDB, React) with **sub-100 ms** median API latency, JWT session management, and Redis rate limiting (100 req/min per user) to maintain stability under burst traffic.